

Project 2 & 3: Exploratory & Advanced Data Analytics

🔗 **Parent Page:** SQL Masterclass: Data Engineering & Analytics

> 💡 **Project Objective:**
> Conduct a comprehensive Exploratory Data Analysis (EDA) to profile raw data and extract actionable business metrics, progressing into advanced business intelligence queries leveraging complex multi-row SQL capabilities for time-series and cohort tracking.

https://github.com/Youssef-Hany1/sql-advanced-business-analytics

🧠 Key Analytical Pillars

- **Database & Dimension Exploration:** Querying the `INFORMATION_SCHEMA` to understand database structure, and identifying unique values across key categories.
- **Magnitude & Ranking Analysis:** Aggregating measures by dimensions (e.g., Total Sales by Country) and utilizing sorting/ranking to identify Top N and Bottom N performers.
- **Change Over Time & Cumulative Analysis:** Tracking trends and seasonality across years/months, and calculating running totals and moving averages to monitor overarching business growth.
- **Performance & Part-to-Whole Analysis:** Conducting Year-over-Year (YoY) comparisons, evaluating current performance against averages/extremes, and finding percentage contributions of individual categories to the overall total.
- **Data Segmentation:** Grouping customers into categories (VIP, Regular, New) based on spending behavior and lifespan using conditional logic.

🛠️ Technical Implementation

- **Core Functions:** Aggregate Functions (`SUM`, `MIN`, `MAX`, `AVG`, `COUNT`), `GROUP BY`, `TOP N`.
- **Advanced Queries:** Common Table Expressions (CTEs), Subqueries, `CASE WHEN` statements, Set Operators (`UNION`, `EXCEPT`).
- **Window Functions:** `OVER()`, `PARTITION BY`, Value Functions (`LEAD`, `LAG`, `FIRST_VALUE`, `LAST_VALUE`).
- **Ranking Functions:** `ROW_NUMBER`, `RANK`, `DENSE_RANK`.

Tables

Projects

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Exploratory Data Analysis Project (EDA)	1
SQL Advanced Data Analytics Project	1

Project Tasks

↗ Projects	Aa Task	☑ Check
Exploratory Data Analysis Project (EDA)	Database Exploration	☑
Exploratory Data Analysis Project (EDA)	Dimension Exploration	☑
Exploratory Data Analysis Project (EDA)	Data Exploration	☑
Exploratory Data Analysis Project (EDA)	Measure Exploration	☑
Exploratory Data Analysis Project (EDA)	Magnitude Exploration	☑
Exploratory Data Analysis Project (EDA)	Ranking Analysis	☑
SQL Advanced Data Analytics Project	Change Over Time Analysis	☑
SQL Advanced Data Analytics Project	Cumulative Analysis	☑
SQL Advanced Data Analytics Project	Performance Analysis	☑
SQL Advanced Data Analytics Project	Part to Whole Analysis	☑
SQL Advanced Data Analytics Project	Data Segmentation	☑
SQL Advanced Data Analytics Project	Build Customers Report	☑
SQL Advanced Data Analytics Project	Build Products Report	☑
SQL Advanced Data Analytics Project	Document Your Work in GIT	☑